

CO5 AT-3

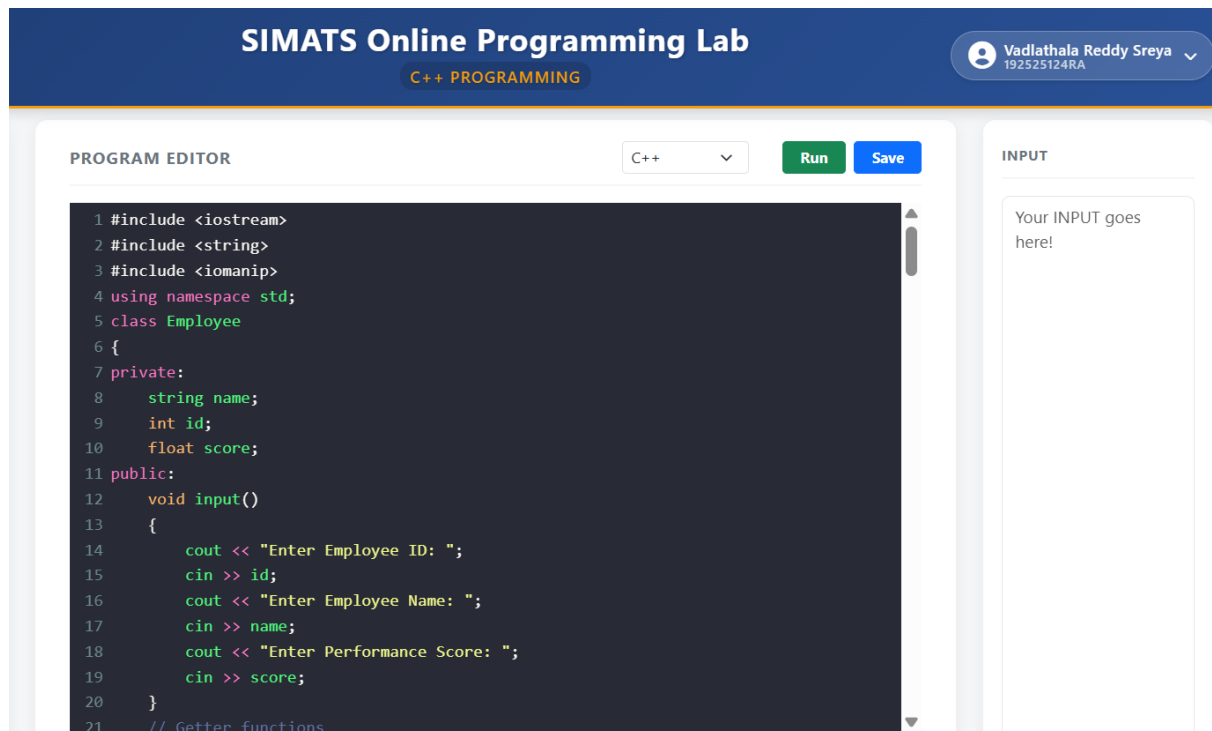
DSA0103-Object Oriented Programming using C++

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Regno: 192525124

1. A multinational company wants to develop an Employee Performance Analytics System to evaluate employee productivity across multiple departments, where each department has a varying number of employees and performance scores. Design a C++ program using functions and object-oriented concepts such as classes, objects, access specifiers, member functions, static members, and arrays of objects to represent and manage employee data effectively. The program should compute the top performer in each department, identify the overall best employee, and calculate the average performance score per department. Additionally, the solution must handle variable department sizes, edge cases such as no employees, equal scores, or all low-performing employees, and ensure efficient comparisons, optimized time complexity, and modular program design.

Code:



The screenshot displays the SIMATS Online Programming Lab interface. At the top, a dark blue header contains the text "SIMATS Online Programming Lab" and "C++ PROGRAMMING". On the right side of the header, there is a user profile icon and the name "Vadlathala Reddy Sreya" with the ID "192525124RA". Below the header, the interface is divided into three main sections. The left section is the "PROGRAM EDITOR", which has a dropdown menu set to "C++" and buttons for "Run" and "Save". The editor contains the following C++ code:

```
1 #include <iostream>
2 #include <string>
3 #include <iomanip>
4 using namespace std;
5 class Employee
6 {
7 private:
8     string name;
9     int id;
10    float score;
11 public:
12    void input()
13    {
14        cout << "Enter Employee ID: ";
15        cin >> id;
16        cout << "Enter Employee Name: ";
17        cin >> name;
18        cout << "Enter Performance Score: ";
19        cin >> score;
20    }
21    // Getter functions
```

The right section is labeled "INPUT" and contains a text area with the placeholder text "Your INPUT goes here!".

SIMATS Online Programming Lab

C++ PROGRAMMING



Vadlathala Reddy Sreya
192525124RA

PROGRAM EDITOR

C++

Run

Save

```
22     float getScore() const
23     {
24         return score;
25     }
26     string getName() const
27     {
28         return name;
29     }
30     int getId() const
31     {
32         return id;
33     }
34 };
35 class Department
36 {
37 private:
38     string departmentName;
39     Employee employees[100];
40     int employeeCount;
41     // Static member
```

INPUT

Your INPUT goes
here!

SIMATS Online Programming Lab

C++ PROGRAMMING



Vadlathala Reddy Sreya
192525124RA

PROGRAM EDITOR

C++

Run

Save

```
42     static int totalEmployees;
43 public:
44     // Constructor
45     Department()
46     {
47         employeeCount = 0;
48     }
49     // Input department details
50     void inputDepartment()
51     {
52         cout << "\nEnter Department Name: ";
53         cin >> departmentName;
54         cout << "Enter Number of Employees: ";
55         cin >> employeeCount;
56         if (employeeCount < 0)
57             employeeCount = 0;
58         if (employeeCount > 100)
59             employeeCount = 100;
60         totalEmployees += employeeCount;
61         for (int i = 0; i < employeeCount; i++)
62         {
```

INPUT

Your INPUT goes
here!

SIMATS Online Programming Lab

C++ PROGRAMMING

Vadlathala Reddy Sreya
192525124RA

PROGRAM EDITOR

C++

Run

Save

```
63         cout << "\nEmployee " << i + 1 << endl;
64         employees[i].input();
65     }
66 }
67 // Find top performer
68 void topPerformer()
69 {
70     if (employeeCount == 0)
71     {
72         cout << "\n" << departmentName
73             << ": No employees available.\n";
74         return;
75     }
76     float highest = employees[0].getScore();
77     int index = 0;
78     for (int i = 1; i < employeeCount; i++)
79     {
80         if (employees[i].getScore() > highest)
81         {
82             highest = employees[i].getScore();
83             index = i;
```

INPUT

Your INPUT goes
here!

SIMATS Online Programming Lab

C++ PROGRAMMING

Vadlathala Reddy Sreya
192525124RA

PROGRAM EDITOR

C++

Run

Save

```
84     }
85 }
86 cout << "\nDepartment: " << departmentName;
87 cout << "\nTop Performer: " << employees[index].getName();
88 cout << "\nEmployee ID: " << employees[index].getId();
89 cout << "\nScore: " << highest << endl;
90 }
91 // Calculate average performance
92 float averageScore()
93 {
94     if (employeeCount == 0)
95         return 0;
96     float sum = 0;
97     for (int i = 0; i < employeeCount; i++)
98     {
99         sum += employees[i].getScore();
100     }
101     return sum / employeeCount;
102 }
103 string getDepartmentName()
104 {
```

INPUT

Your INPUT goes
here!

SIMATS Online Programming Lab

C++ PROGRAMMING



Vadlathala Reddy Sreya
192525124RA

PROGRAM EDITOR

C++

Run

Save

```
105     return departmentName;
106 }
107
108 int getEmployeeCount()
109 {
110     return employeeCount;
111 }
112 // Static function
113 static int getTotalEmployees()
114 {
115     return totalEmployees;
116 }
117 // Get employee with highest score
118 Employee getBestEmployee()
119 {
120     int index = 0;
121     for (int i = 1; i < employeeCount; i++)
122     {
123         if (employees[i].getScore() >
124             employees[index].getScore())
125         {
```

INPUT

Your INPUT goes
here!

SIMATS Online Programming Lab

C++ PROGRAMMING



Vadlathala Reddy Sreya
192525124RA

PROGRAM EDITOR

C++

Run

Save

```
126         index = i;
127     }
128 }
129 return employees[index];
130 }
131 };
132 // Definition of static member
133 int Department::totalEmployees = 0;
134 int main()
135 {
136     int n;
137     cout << "=====\n";
138     cout << " EMPLOYEE PERFORMANCE ANALYTICS SYSTEM\n";
139     cout << "=====\n";
140     cout << "\nEnter number of departments: ";
141     cin >> n;
142     if (n <= 0)
143     {
144         cout << "No departments available.\n";
145         return 0;
146     }
```

INPUT

Your INPUT goes
here!

SIMATS Online Programming Lab

C++ PROGRAMMING

Vadlathala Reddy Sreya
192525124RA

PROGRAM EDITOR

C++

Run

Save

```
147     if (n > 20)
148         n = 20;
149     Department departments[20];
150     // Input departments
151     for (int i = 0; i < n; i++)
152     {
153         cout << "\n===== Department "
154              << i + 1 << " =====\n";
155
156         departments[i].inputDepartment();
157     }
158     cout << "\n\n===== RESULTS =====\n";
159     float overallHighest = -1;
160     string overallName;
161     int overallId = -1;
162     string overallDepartment;
163     // Display department results
164     for (int i = 0; i < n; i++)
165     {
166         departments[i].topPerformer();
167         cout << fixed << setprecision(2);
```

INPUT

Your INPUT goes
here!

SIMATS Online Programming Lab

C++ PROGRAMMING

Vadlathala Reddy Sreya
192525124RA

PROGRAM EDITOR

C++

Run

Save

```
168     cout << "Average Score: "
169           << departments[i].averageScore() << endl;
170     // Find overall best employee
171     if (departments[i].getEmployeeCount() > 0)
172     {
173         Employee best = departments[i].getBestEmployee();
174         if (best.getScore() > overallHighest)
175         {
176             overallHighest = best.getScore();
177             overallName = best.getName();
178             overallId = best.getId();
179             overallDepartment =
180                 departments[i].getDepartmentName();
181         }
182     }
183 }
184 cout << "\n===== \n";
185 cout << "          OVERALL BEST EMPLOYEE\n";
186 cout << "===== \n";
187 if (overallHighest == -1)
188 {
```

INPUT

Your INPUT goes
here!

SIMATS Online Programming Lab

C++ PROGRAMMING

Vadlathala Reddy Sreya
192525124RA

```
189     cout << "No employees available.\n";
190 }
191 else
192 {
193     cout << "Name      : " << overallName << endl;
194     cout << "Employee ID: " << overallId << endl;
195     cout << "Department : " << overallDepartment << endl;
196     cout << "Score      : " << overallHighest << endl;
197 }
198 cout << "\nTotal Employees: "
199     << Department::getTotalEmployees() << endl;
200
201 return 0;
202 }
```

Input:

Vadlathala Reddy Sreya
192525124RA

```
3
HR
3
101
Ravi
85
102
Arun
92
103
Kiran
78
IT
2
201
Priya
95
202
Rahul
88
Finance
2
301
Anu
90
302
Sita
90
```

Output:

SIMATS Online Programming Lab

C++ PROGRAMMING

Vadlathala Reddy Sreya

192525124RA

204

205

OUTPUT

EMPLOYEE PERFORMANCE ANALYTICS SYSTEM

=====

Enter number of departments:

===== Department 1 =====

Enter Department Name: Enter Number of Employees:

Employee 1

Enter Employee ID: Enter Employee Name: Enter Performance Score:

Employee 2

88

Finance

2

301

Anu

90

302

Sita

90

SIMATS Online Programming Lab

C++ PROGRAMMING

Vadlathala Reddy Sreya

192525124RA

204

205

OUTPUT

Enter Employee ID: Enter Employee Name: Enter Performance Score:

Employee 3

Enter Employee ID: Enter Employee Name: Enter Performance Score:

===== Department 2 =====

Enter Department Name: Enter Number of Employees:

Employee 1

Enter Employee ID: Enter Employee Name: Enter Performance Score:

Employee 2

Enter Employee ID: Enter Employee Name: Enter Performance Score:

88

Finance

2

301

Anu

90

302

Sita

90

SIMATS Online Programming Lab

C++ PROGRAMMING

Vadlathala Reddy Sreya

192525124RA

204

205

OUTPUT

===== Department 3 =====

Enter Department Name: Enter Number of Employees:

Employee 1

Enter Employee ID: Enter Employee Name: Enter Performance Score:

Employee 2

Enter Employee ID: Enter Employee Name: Enter Performance Score:

===== RESULTS =====

88

Finance

2

301

Anu

90

302

Sita

90

SIMATS Online Programming Lab

C++ PROGRAMMING

Vadlathala Reddy Sreya

192525124RA

204

205

OUTPUT

Department: HR

Top Performer: Arun

Employee ID: 102

Score: 92

Average Score: 85.00

Department: IT

Top Performer: Priya

Employee ID: 201

Score: 95.00

88

Finance

2

301

Anu

90

302

Sita

90

SIMATS Online Programming Lab

C++ PROGRAMMING

Vadlathala Reddy Sreya

192525124RA

204

205

OUTPUT

Average Score: 91.50

Department: Finance

Top Performer: Anu

Employee ID: 301

Score: 90.00

Average Score: 90.00

=====

OVERALL BEST EMPLOYEE

88

Finance

2

301

Anu

90

302

Sita

90

SIMATS Online Programming Lab

C++ PROGRAMMING

Vadlathala Reddy Sreya

192525124RA

204

205

OUTPUT

=====

OVERALL BEST EMPLOYEE

=====

Name : Priya

Employee ID: 201

Department : IT

Score : 95.00

Total Employees: 7

88

Finance

2

301

Anu

90

302

Sita

90

2. A logistics company wants to develop a Fleet Management System to monitor multiple vehicles across different routes, where each route has a varying number of vehicles and trip records. Design a C++ program using functions and object-oriented concepts such as classes, objects, access specifiers, member functions, static members, and arrays of objects to represent vehicle data efficiently. The program should compute the highest distance covered vehicle on each route, identify the overall most efficient vehicle, and calculate the average distance per route. Additionally, the solution must handle variable fleet sizes, edge cases such as no vehicles, identical distances, or zero trips, and ensure efficient data processing, modular design, and optimized comparisons.

Code:

SIMATS Online Programming Lab

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192525124RA

C++ PROGRAMMING

```
1 #include <iostream>
2 #include <string>
3 #include <iomanip>
4 using namespace std;
5
6 class Vehicle
7 {
8 private:
9     int vehicleId;
10    string vehicleName;
11    float distance;
12
13 public:
14    // Input vehicle details
15    void input()
16    {
17        cout << "Enter Vehicle ID: ";
18        cin >> vehicleId;
19
20        cout << "Enter Vehicle Name: ";
```

Your INPUT goes here!

SIMATS Online Programming Lab

C++ PROGRAMMING



Vadlathala Reddy Sreya
192525124RA

```
21     cin >> vehicleName;
22
23     cout << "Enter Distance Covered (km): ";
24     cin >> distance;
25
26     if (distance < 0)
27         distance = 0;
28 }
29
30 // Getter functions
31 float getDistance() const
32 {
33     return distance;
34 }
35
36 string getName() const
37 {
38     return vehicleName;
39 }
40
```

Your INPUT goes
here!

SIMATS Online Programming Lab

C++ PROGRAMMING



Vadlathala Reddy Sreya
192525124RA

```
41 int getId() const
42 {
43     return vehicleId;
44 }
45 };
46
47 class Route
48 {
49 private:
50     string routeName;
51     Vehicle vehicles[100];
52     int vehicleCount;
53
54     // Static member
55     static int totalVehicles;
56
57 public:
58     // Constructor
59     Route()
60     {
```

Your INPUT goes
here!

SIMATS Online Programming Lab

C++ PROGRAMMING



Vadlathala Reddy Sreya
192525124RA



```
61     vehicleCount = 0;
62 }
63
64 // Input route details
65 void inputRoute()
66 {
67     cout << "\nEnter Route Name: ";
68     cin >> routeName;
69
70     cout << "Enter Number of Vehicles: ";
71     cin >> vehicleCount;
72
73     if (vehicleCount < 0)
74         vehicleCount = 0;
75
76     if (vehicleCount > 100)
77         vehicleCount = 100;
78
79     totalVehicles += vehicleCount;
80
```

Your INPUT goes
here!

SIMATS Online Programming Lab

C++ PROGRAMMING



Vadlathala Reddy Sreya
192525124RA



```
81     for (int i = 0; i < vehicleCount; i++)
82     {
83         cout << "\nVehicle " << i + 1 << endl;
84         vehicles[i].input();
85     }
86 }
87
88 // Find vehicle with highest distance
89 void highestDistanceVehicle()
90 {
91     if (vehicleCount == 0)
92     {
93         cout << "\n" << routeName
94             << ": No vehicles available.\n";
95         return;
96     }
97
98     float highest = vehicles[0].getDistance();
99     int index = 0;
100
```

Your INPUT goes
here!

SIMATS Online Programming Lab

C++ PROGRAMMING



Vadlathala Reddy Sreya
192525124RA

```
101     for (int i = 1; i < vehicleCount; i++)
102     {
103         if (vehicles[i].getDistance() > highest)
104         {
105             highest = vehicles[i].getDistance();
106             index = i;
107         }
108     }
109
110     cout << "\nRoute: " << routeName;
111     cout << "\nHighest Distance Vehicle: "
112           << vehicles[index].getName();
113     cout << "\nVehicle ID: "
114           << vehicles[index].getId();
115     cout << "\nDistance: "
116           << highest << " km\n";
117 }
118
119 // Calculate average distance
120 float averageDistance()
```

Your INPUT goes
here!

SIMATS Online Programming Lab

C++ PROGRAMMING



Vadlathala Reddy Sreya
192525124RA

```
121 {
122     if (vehicleCount == 0)
123         return 0;
124
125     float sum = 0;
126
127     for (int i = 0; i < vehicleCount; i++)
128     {
129         sum += vehicles[i].getDistance();
130     }
131
132     return sum / vehicleCount;
133 }
134
135 // Getters
136 string getRouteName()
137 {
138     return routeName;
139 }
140
```

Your INPUT goes
here!



```
141 int getVehicleCount()
142 {
143     return vehicleCount;
144 }
145
146 // Get most efficient vehicle
147 Vehicle getMostEfficientVehicle()
148 {
149     int index = 0;
150
151     for (int i = 1; i < vehicleCount; i++)
152     {
153         if (vehicles[i].getDistance() >
154             vehicles[index].getDistance())
155         {
156             index = i;
157         }
158     }
159
160     return vehicles[index];
```

Your INPUT goes
here!



```
161 }
162
163 // Static function
164 static int getTotalVehicles()
165 {
166     return totalVehicles;
167 }
168 };
169
170 // Definition of static member
171 int Route::totalVehicles = 0;
172
173 int main()
174 {
175     int n;
176
177     cout << "=====\n";
178     cout << "        FLEET MANAGEMENT SYSTEM\n";
179     cout << "=====\n";
180
```

Your INPUT goes
here!

SIMATS Online Programming Lab

C++ PROGRAMMING



Vadlathala Reddy Sreya
192525124RA

```
181 cout << "\nEnter number of routes: ";
182 cin >> n;
183
184 if (n <= 0)
185 {
186     cout << "No routes available.\n";
187     return 0;
188 }
189
190 if (n > 20)
191     n = 20;
192
193 Route routes[20];
194
195 // Input route information
196 for (int i = 0; i < n; i++)
197 {
198     cout << "\n===== Route "
199           << i + 1 << " =====\n";
200
```

Your INPUT goes
here!

SIMATS Online Programming Lab

C++ PROGRAMMING



Vadlathala Reddy Sreya
192525124RA

```
201     routes[i].inputRoute();
202 }
203
204 cout << "\n\n===== FLEET RESULTS =====\n";
205
206 float overallHighest = -1;
207 string overallVehicle;
208 string overallRoute;
209 int overallId = -1;
210
211 // Process all routes
212 for (int i = 0; i < n; i++)
213 {
214     routes[i].highestDistanceVehicle();
215
216     cout << fixed << setprecision(2);
217
218     cout << "Average Distance: "
219           << routes[i].averageDistance()
220           << " km\n";

```

Your INPUT goes
here!

SIMATS Online Programming Lab

C++ PROGRAMMING



Vadlathala Reddy Sreya
192525124RA

```
221
222     // Find overall most efficient vehicle
223     if (routes[i].getVehicleCount() > 0)
224     {
225         Vehicle best =
226             routes[i].getMostEfficientVehicle();
227
228         if (best.getDistance() > overallHighest)
229         {
230             overallHighest = best.getDistance();
231             overallVehicle = best.getName();
232             overallId = best.getId();
233             overallRoute = routes[i].getRouteName();
234         }
235     }
236 }
237
238 cout << "\n===== \n";
239 cout << "    OVERALL MOST EFFICIENT VEHICLE \n";
240 cout << "===== \n";
```

Your INPUT goes
here!

SIMATS Online Programming Lab

C++ PROGRAMMING



Vadlathala Reddy Sreya
192525124RA

```
241
242     if (overallHighest == -1)
243     {
244         cout << "No vehicles available. \n";
245     }
246     else
247     {
248         cout << "Vehicle Name: "
249             << overallVehicle << endl;
250
251         cout << "Vehicle ID : "
252             << overallId << endl;
253
254         cout << "Route      : "
255             << overallRoute << endl;
256
257         cout << "Distance   : "
258             << overallHighest << " km \n";
259     }
260
```

Your INPUT goes
here!

SIMATS Online Programming Lab

C++ PROGRAMMING



Vadlathala Reddy Sreya
192525124RA

```
261     cout << "\nTotal Vehicles: "
262         << Route::getTotalVehicles() << endl;
263
264     return 0;
265 }
```

Input:

Vadlathala Reddy Sreya
192525124RA

3

Chennai-Hyderabad

3

101

Truck1

450

102

Truck2

520

103

Truck3

480

Bangalore-Chennai

2

201

Truck4

600

202

Truck5

550

Delhi-Mumbai

2

301

Truck6

700

302

Truck7

650

Output:

SIMATS Online Programming Lab

C++ PROGRAMMING

Vadlathala Reddy Sreya
192525124RA

266

267

OUTPUT

```
=====
FLEET MANAGEMENT SYSTEM
=====

Enter number of routes:
===== Route 1 =====

Enter Route Name: Enter Number of Vehicles:
Vehicle 1
Enter Vehicle ID: Enter Vehicle Name: Enter Distance Covered
(km):
```


SIMATS Online Programming Lab

C++ PROGRAMMING



Vadlathala Reddy Sreya
192525124RA

```
266  
267
```

OUTPUT

```
Vehicle 2  
Enter Vehicle ID: Enter Vehicle Name: Enter Distance Covered  
(km):  
Vehicle 3  
Enter Vehicle ID: Enter Vehicle Name: Enter Distance Covered  
(km):  
===== Route 2 =====  
  
Enter Route Name: Enter Number of Vehicles:  
Vehicle 1
```

SIMATS Online Programming Lab

C++ PROGRAMMING



Vadlathala Reddy Sreya
192525124RA

```
266  
267 |
```

OUTPUT

```
Enter Vehicle ID: Enter Vehicle Name: Enter Distance Covered  
(km):  
Vehicle 2  
Enter Vehicle ID: Enter Vehicle Name: Enter Distance Covered  
(km):  
===== Route 3 =====  
  
Enter Route Name: Enter Number of Vehicles:  
Vehicle 1  
Enter Vehicle ID: Enter Vehicle Name: Enter Distance Covered
```

SIMATS Online Programming Lab

C++ PROGRAMMING



Vadlathala Reddy Sreya
192525124RA

```
266  
267 |
```

OUTPUT

```
(km):  
Vehicle 2  
Enter Vehicle ID: Enter Vehicle Name: Enter Distance Covered  
(km):  
  
===== FLEET RESULTS =====  
  
Route: Chennai-Hyderabad  
Highest Distance Vehicle: Truck2  
Vehicle ID: 102  
Distance: 520 km
```

SIMATS Online Programming Lab

C++ PROGRAMMING



Vadlathala Reddy Sreya
192525124RA



```
266  
267 |
```

OUTPUT

```
Distance: 520 km  
Average Distance: 483.33 km  
  
Route: Bangalore-Chennai  
Highest Distance Vehicle: Truck4  
Vehicle ID: 201  
Distance: 600.00 km  
Average Distance: 575.00 km  
  
Route: Delhi-Mumbai
```

SIMATS Online Programming Lab

C++ PROGRAMMING



Vadlathala Reddy Sreya
192525124RA



```
266  
267
```

OUTPUT

```
Highest Distance Vehicle: Truck6  
Vehicle ID: 301  
Distance: 700.00 km  
Average Distance: 675.00 km
```

```
=====
```

```
OVERALL MOST EFFICIENT VEHICLE
```

```
=====
```

```
Vehicle Name: Truck6  
Vehicle ID : 301
```

SIMATS Online Programming Lab

C++ PROGRAMMING



Vadlathala Reddy Sreya
192525124RA



```
266  
267
```

OUTPUT

```
=====
```

```
OVERALL MOST EFFICIENT VEHICLE
```

```
=====
```

```
Vehicle Name: Truck6  
Vehicle ID : 301  
Route      : Delhi-Mumbai  
Distance   : 700.00 km  
  
Total Vehicles: 7
```